# STEVEN YANG

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## Education

**University of Toronto** **Sep. 2024 – May 2028 (Expected)**

- *Honours Bachelor of Science*, Data Science Specialist, Computer Science Major
- cGPA: 4.0/4.0
- **Awards:** Dean's List Scholar (2025, 2026) and Woodsworth College Scholarship Fund (2026)
- Relevant coursework: Software Design, Software Tools and System Programming, Data Structures and Analysis, Data Science, Probability and Statistics, Linear Algebra, Multivariable Calculus

## Research

**Agentic Languages Evaluations** **Sep. 2026 – Present**

*University of Toronto*

- Designed **AgentLangBench** to test whether languages built for AI agents help them produce correct, **formally verified**, and faster code. Screened 42 languages and tools using smoke tests to select benchmark candidates.
- Built an evaluation pipeline that runs coding agents in isolated **Docker** containers and automatically grades their submissions using hidden tests and language-specific verifiers.
- Select tasks with scalable inputs and verifiable outputs from cross-language benchmark suites. Compare runtime against C and Python on identical inputs after checking output **correctness** and matching algorithmic complexity.

## Experience

**Research and Development Intern** **May 2026 – Aug. 2026**

*NSFOCUS* *Beijing, China*

- Built a Python pipeline using the **Tavily** and **FOFA** APIs to find suspicious domains and social media accounts, remove duplicates, and score each result by risk.
- Prototyped a **DeepSeek** workflow to match threat events to client profiles, remove duplicates, and determine which events to deliver.
- Wrote a research report analyzing the **architecture harness** and workflow of an AI coding agent.

## Projects

**NHANES Supplement Use Analysis** *Python, pandas, NumPy, matplotlib, scikit-learn* **2026**

- Worked with a teammate to analyze how supplement use varied with diet and demographic characteristics in the 2021–2023 NHANES survey.
- Cleaned and preprocessed linked survey datasets and engineered features for modeling.
- Modeled supplement use with statistical and **machine learning** methods and visualized predictive factors and model performance.

**Question Bank Platform** *Java, OpenAI API* **2026**

- Built a full-stack Java application to extract and structure math exam questions from PDFs into a searchable digital database.
- Combined **OCR** and OpenAI vision models to extract question text, diagrams, and metadata from page images.
- Designed an asynchronous pipeline to segment PDFs, store results as JSON, and flag low-confidence extractions for review.

**GroupFlow** *Java, Swing, MongoDB, JUnit* **2025**

- Designed and implemented a collaborative task and group management application using **Clean Architecture**, separating entities, use cases, and UI layers.
- Built core features including group creation, role-based permissions, task assignment, and progress tracking.
- Used interfaces to isolate components for unit testing with **JUnit**.
- Reviewed teammates' code and integrated changes through Git pull requests.

## Technical Skills

- **Languages:** Python, Java, C, JavaScript, Rust, HTML/CSS, Verus, Dafny
- **Developer Tools:** Codex, Claude Code, Git/GitHub, Linux, VS Code, IntelliJ, Jupyter, LaTeX, PostgreSQL, Docker
- **Frameworks/Libraries:** NumPy, pandas, matplotlib, scikit-learn, SciPy, statsmodels, TensorFlow, PyTorch